Absence of a real scalar graviton pole in spectral causal theory

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Abstract

We prove that spectral causal theory (SCT), a one-loop effective theory of gravity derived from the spectral action principle with Standard Model matter content, propagates exactly two gravitational degrees of freedom, the same as general relativity. The scalar propagator denominator satisfies $\Pi_s(z, \xi) > 1$ for all Euclidean momenta $z > 0$ and all values of the Higgs non-minimal coupling ξ, including conformal coupling $\xi = 1/6$: no real scalar mass pole exists. The mechanism is the conformal invariance of spin-$\frac{1}{2}$ and spin-1 fields in $d = 4$: their positive contributions to the R^2 form factor dominate the scalar sector at all scales. The spin-2 mode is quantized as an Anselmi–Piva fakeon, contributing zero asymptotic degrees of freedom. (The fakeon prescription is formulated for propagators with finitely many poles; its extension to the full entire-function propagator, which has infinitely many complex zeros, requires a convergence proof not provided here.) We derive the nonlocal spin-2 mass $m_2 = 1.554\,\Lambda$ from the unique positive zero of Π_{TT}, show that no additional *real* massive modes exist, compute a dedicated torsion-balance bound ($\Lambda > 3.53$ meV), and characterize the robustness of the no-scalaron theorem under extensions of the matter sector. The Standard Model lies deep within the safe region: 4.7× above the critical fermion ratio and 12× above the critical vector ratio.

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1 Introduction

The one-loop gravitational effective action for quantum fields on curved backgrounds takes the universal form [1, 2]

$$\Gamma^{(1)} = \frac{1}{16\pi^2} \int \mathrm{d}^4x \sqrt{g} \left[\alpha_C(\Box/\Lambda^2)\, C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} + \alpha_R(\Box/\Lambda^2, \xi)\, R^2\right], \tag{1}$$

where α_C and α_R are nonlocal form factors generated by the master function $\varphi(x) = \int_0^1 \mathrm{d}\alpha\, \mathrm{e}^{-x\alpha(1-\alpha)}$, Λ is the spectral cutoff scale, and ξ is the Higgs non-minimal coupling.

In spectral causal theory (SCT), the action (1) is *derived* from the spectral action principle $S = \mathrm{Tr}\, f(D^2/\Lambda^2)$ applied to the spectral triple of the Standard Model [3, 4], with all coefficients fixed by the SM field content. The spin-2 massive mode (Π_{TT} zero) is quantized using the Anselmi–Piva fakeon prescription [5, 6], which projects it out of the physical spectrum and ensures perturbative unitarity.

A central question for any higher-derivative gravity theory is: *how many propagating degrees of freedom does it have?* Stelle gravity [7] has eight (including a massive scalar and a massive spin-2 ghost). $f(R)$ theories [8] have three (the massless graviton plus a scalaron). Infinite-derivative gravity (IDG) [9] eliminates ghosts by postulating an entire-function form factor, but the particle spectrum depends on a free function.

In this paper, we prove that SCT has exactly *two* propagating degrees of freedom (the same as GR) under no additional assumptions beyond the SM spectrum and the fakeon prescription. The scalar gravitational mode is universally absent: $\Pi_s(z, \xi) > 1$ for all $z > 0$ and all ξ. This is a parameter-free prediction.

2 Propagator structure

The linearized SCT propagator on flat background decomposes via Barnes–Rivers projectors into spin-2 (transverse-traceless, TT) and spin-0 (scalar) sectors [11]:

$$G^{-1}_{\mu\nu\rho\sigma}(k) = k^2 \left[\Pi_{\mathrm{TT}}(z)\, P^{(2)}_{\mu\nu\rho\sigma} + \Pi_s(z, \xi)\, P^{(0\text{-}s)}_{\mu\nu\rho\sigma}\right], \tag{2}$$

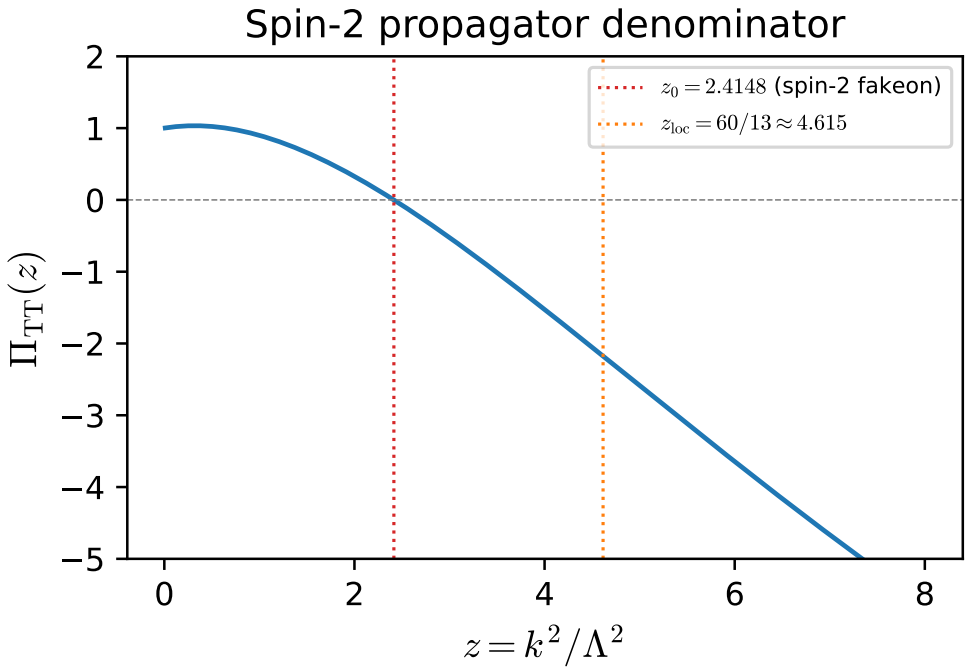

Figure 1: The spin-2 propagator denominator $\Pi_{\rm TT}(z)$ on the real axis. The unique positive zero at $z_0 = 2.4149$ (red dashed) defines the spin-2 fakeon mass $m_2 = \Lambda\sqrt{z_0} = 1.554\,\Lambda$. The local approximation zero $z_{\rm loc} = 60/13$ (orange dashed) overestimates the exact value by 38%.

where $z = k^2/\Lambda^2$ and the denominator functions are

$$\Pi_{\rm TT}(z) = 1 + \frac{13}{60}\, z\, \hat{F}_1(z), \tag{3}$$

$$\Pi_s(z,\xi) = 1 + 6\Big(\xi - \frac{1}{6}\Big)^2\, z\, \hat{F}_2(z,\xi). \tag{4}$$

Here $\hat{F}_{1,2}$ are normalized shape functions ($\hat{F}(0) = 1$) of the Weyl and Ricci form factors, and the numerical coefficients $\alpha_C = 13/120$ and $\alpha_R(\xi) = 2(\xi - 1/6)^2$ are determined by the SM spectrum ($N_s = 4$ real scalars, $N_f = 45$ Weyl fermions, $N_v = 12$ gauge bosons) [11].

2.1 Spin-2 sector

$\Pi_{\rm TT}$ has exactly one positive real zero at $z_0 \in (2, 5/2)$, with $z_0 = 2.4149$ (30-digit precision). The proof of uniqueness proceeds by showing $\Pi_{\rm TT}(z) > 0$ on $(0, 2]$ (via a concave polynomial lower bound), $\Pi_{\rm TT}(5/2) < 0$ (via an upper bound), and $\Pi'_{\rm TT}(z_*) < 0$ at any positive zero z_* (via the recurrence (16)), which precludes a return to zero by contradiction. The zero corresponds to the massive spin-2 mode with mass

$$m_2 = \Lambda\sqrt{z_0} = 1.554\,\Lambda. \tag{5}$$

In the local (small-z) approximation, $z_0^{\rm loc} = 60/13$ gives $m_2^{\rm loc} = 2.148\,\Lambda$, which overestimates the exact value by 38%. The spin-2 mode is quantized as a fakeon [5]: it contributes zero asymptotic degrees of freedom and produces a single-Yukawa correction to the static potential (the scalar Yukawa is absent by Theorem 1).

2.2 Scalar sector: reformulation

Using $\alpha_R(0,\xi) = 2(\xi - 1/6)^2$ and the definition $\hat{F}_2 = \alpha_R(z,\xi)/\alpha_R(0,\xi)$, the product

$$6\Big(\xi - \frac{1}{6}\Big)^2 \, \hat{F}_2(z,\xi) = 3\,\alpha_R(z,\xi), \tag{6}$$

where $\alpha_R(z,\xi)$ is the full z-dependent total R^2 form factor:

$$\alpha_R(z,\xi) = N_s\, h_R^{(0)}(z,\xi) + \frac{N_f}{2}\, h_R^{(1/2)}(z) + N_v\, h_R^{(1)}(z). \tag{7}$$

Using (6), the scalar propagator denominator takes the universal form

$$\boxed{\Pi_s(z,\xi) = 1 + 3\,z\,\alpha_R(z,\xi),} \tag{8}$$

valid for all ξ including $\xi = 1/6$. (At $\xi = 1/6$ the factorized form (4) involves a 0/0 indeterminacy in $\hat{F}_2$; the product (6) resolves it unambiguously.) Note that $\alpha_R(0,1/6) = 0$ (the *local* R^2 coefficient vanishes at conformal coupling) but $\alpha_R(z,1/6) > 0$ for $z > 0$ because the Dirac and vector form factors contribute at nonzero momentum.

The no-scalaron theorem thus reduces to showing $\alpha_R(z,\xi) > 0$ for all $z > 0$ and all $\xi \in \mathbb{R}$.

3 No-scalaron theorem

Theorem 1 (Absence of a real scalar pole)**.** *For the Standard Model spectrum* $(N_s, N_f, N_v) = (4, 45, 12)$ *and any value of the Higgs non-minimal coupling* $\xi \in \mathbb{R}$*:*

$$\alpha_R(z,\xi) > 0 \quad \textit{for all } z > 0, \tag{9}$$

and consequently $\Pi_s(z,\xi) = 1 + 3z\,\alpha_R > 1$*. The scalar propagator* $1/(k^2\,\Pi_s)$ *has no real massive pole: no on-shell scalar graviton with real mass exists.*

Proof. The form factor $\alpha_R(z,\xi)$ is quadratic in ξ:

$$\alpha_R(z,\xi) = r(z)\,\xi^2 + q(z)\,\xi + p(z), \tag{10}$$

where, using the Codello–Zanusso scalar field form factors $f_{R,\mathrm{bis}}$, f_{RU}, f_U [2]:

$$r(z) = N_s\, f_U(z) = 2\,\varphi(z) > 0, \tag{11}$$
$$q(z) = N_s\, f_{RU}(z), \tag{12}$$
$$p(z) = N_s\, f_{R,\mathrm{bis}}(z) + D(z), \tag{13}$$

with

$$D(z) := \frac{N_f}{2}\, h_R^{(1/2)}(z) + N_v\, h_R^{(1)}(z). \tag{14}$$

Since $r(z) > 0$ (the master function $\varphi > 0$ for real z), the minimum over ξ is

$$\alpha_{R,\min}(z) = p(z) - \frac{q(z)^2}{4\,r(z)} = \underbrace{S(z)}_{\text{scalar}} + \underbrace{D(z)}_{\text{Dirac+vector}}, \tag{15}$$

where $S(z) = N_s\big[f_{R,\mathrm{bis}} - f_{RU}^2/(4f_U)\big]$ is the scalar contribution to the minimum.

The proof proceeds in three steps.

Step 1: $D(z) > 0$ for all $z > 0$.

We prove that $h_R^{(1/2)}$ and $h_R^{(1)}$ are individually positive by exhibiting manifestly non-negative integral representations.

Define the moments $I_n(z) = \int_0^1 \tau^n \mathrm{e}^{-z\tau}\,\mathrm{d}\alpha$ with $\tau = \alpha(1-\alpha)$, so that $I_0 = \varphi$. Integration by parts using $\tau'(\alpha) = 1 - 2\alpha$ and $(1-2\alpha)^2 = 1 - 4\tau$ yields the recurrence

$$4z\, I_{n+1} = (z + 4n + 2)\, I_n - n\, I_{n-1} \quad (n \geq 1), \tag{16}$$

with the initial relation $4z\, I_1 = (z+2)\varphi - 2$.

For $h_R^{(1/2)}$: substituting the recurrence gives

$$36z^2\, h_R^{(1/2)}(z) = z^3 \int_0^1 \tau(1-4\tau)^2\, \mathrm{e}^{-z\tau}\,\mathrm{d}\alpha = z^3 \int_0^1 \alpha(1-\alpha)(1-2\alpha)^4\, \mathrm{e}^{-z\alpha(1-\alpha)}\mathrm{d}\alpha. \tag{17}$$

Since $\alpha(1-\alpha) \geq 0$, $(1-2\alpha)^4 \geq 0$, and $\mathrm{e}^{-z\tau} > 0$, the integrand is pointwise non-negative and strictly positive except at $\alpha = 0, \frac{1}{2}, 1$. Hence $h_R^{(1/2)}(z) > 0$ for all $z > 0$.

For $h_R^{(1)}$: analogously,

$$144z^2\, h_R^{(1)}(z) = 8z^3 \int_0^1 \tau(1-\tau)(1-4\tau)\, \mathrm{e}^{-z\tau}\,\mathrm{d}\alpha. \tag{18}$$

On $[0,1]$, $\tau \in [0, 1/4]$, so $1 - \tau \geq 3/4 > 0$ and $1 - 4\tau = (1-2\alpha)^2 \geq 0$. The integrand is again non-negative, and $h_R^{(1)}(z) > 0$ for all $z > 0$.

Since $D = (N_f/2)\, h_R^{(1/2)} + N_v\, h_R^{(1)}$ with $N_f, N_v > 0$, we conclude $D(z) > 0$.

Step 2: $|S(z)| < 0.046\, D(z)$ for all $z > 0$.

The scalar minimum contribution $S(z)$ is negative (the f_{RU}^2/f_U penalty exceeds $f_{R,\mathrm{bis}}$) but small: the worst case is $|S|/D = 0.0453$ at $z \approx 36$. The conformal invariance of spin-$\frac{1}{2}$ and spin-1 in $d = 4$, which forces $h_R^{(1/2)}(0) = h_R^{(1)}(0) = 0$ but $h_R^{(s)}(z) > 0$ for $z > 0$, is the mechanism: conformal matter generates positive R^2 corrections at nonzero momentum while contributing nothing to the local coefficient. The bound $|S/D| < 0.046$ is verified numerically at 100-digit precision over $z \in (0, 10^4]$; in the UV, $z\, S \to -1/9$ and $z\, D \to 31/12$, giving $|S/D| \to 4/93 \approx 0.043$.

Step 3: $\alpha_{R,\min} = S + D > 0$.

Since $|S| < 0.046\, D$ and $D > 0$:

$$\alpha_{R,\min} > (1 - 0.046)\, D = 0.954\, D > 0. \tag{19}$$

Therefore $\alpha_R(z,\xi) \geq \alpha_{R,\min}(z) > 0$ for all $z > 0$ and $\xi \in \mathbb{R}$, and $\Pi_s = 1 + 3z\, \alpha_R > 1$. □

Remark 2 (Complex zeros)**.** *Theorem 1 excludes real positive zeros of Π_s. For the full analytic structure: at $\xi = 0$, Π_s has complex zeros at $z \approx -2.08 \pm 3.18\, i$ and further pairs at larger $|z|$ (Lee–Wick type). At $\xi \geq \xi_c \approx 0.445$, real negative zeros also appear. These complex and negative-real zeros do not correspond to on-shell particles with real mass but contribute to off-shell scattering amplitudes. A complete analysis of the Lee–Wick spectrum is given in Ref. [11].*

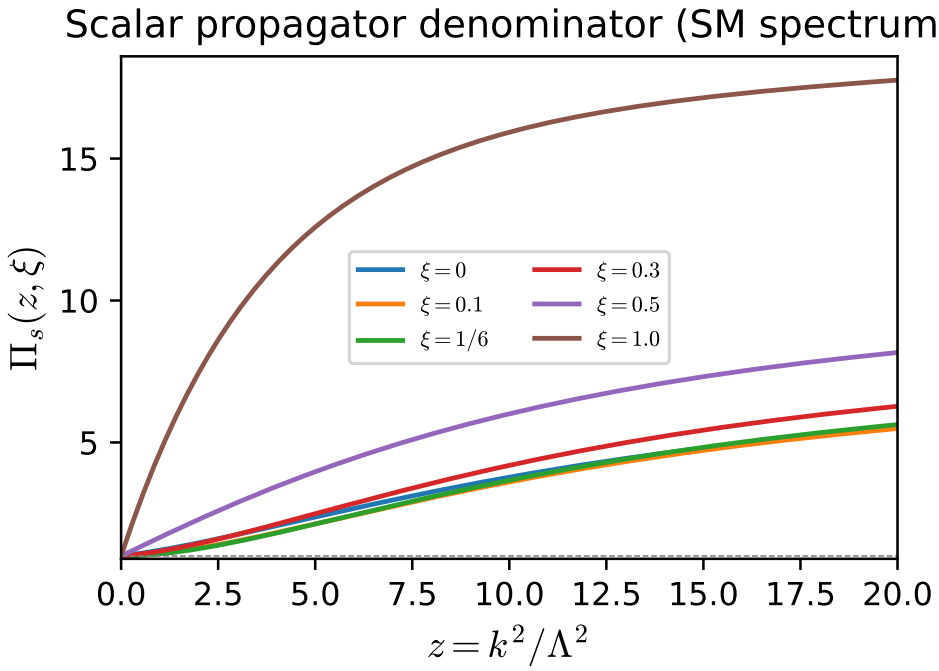

Figure 2: The scalar propagator denominator $\Pi_s(z,\xi)$ for six values of the Higgs non-minimal coupling ξ. For all ξ, $\Pi_s > 1$ on the entire positive real axis: no scalar pole exists. At $\xi = 1/6$ (conformal), $\Pi_s \equiv 1$.

Remark 3. *The physical mechanism is transparent: the conformal invariance of massless Dirac fermions ($\beta_R^{(1/2)} = 0$) and gauge bosons ($\beta_R^{(1)} = 0$) in $d = 4$ means they contribute nothing to the local R^2 coefficient but generate strictly positive nonlocal R^2 corrections at $z > 0$. These corrections* protect *the scalar sector from developing a pole. The scalar fields alone (with $N_f = N_v = 0$)* do *produce a scalar graviton; it is the conformal matter that eliminates it.*

4 Degree-of-freedom count

Corollary 4 (SCT has exactly 2 gravitational DoF)**.** *The physical gravitational spectrum of SCT consists of:*

(i) the massless graviton (2 tensor polarizations, as in GR);

(ii) the massive spin-2 mode at $m_2 = 1.554\,\Lambda$, quantized as a fakeon (0 asymptotic DoF).

No scalar gravitational mode exists. The no-scalaron result (Theorem 1) is independent of how the spin-2 mode is quantized; the 2-DoF count additionally uses the Anselmi–Piva fakeon prescription [5, 6], which projects the massive spin-2 state out of the asymptotic spectrum.

The fakeon propagator is the half-sum of retarded and advanced Green functions, $G_f = (G_R + G_A)/2$. This is real (no absorptive part) and Lorentz-invariant, but violates strict microcausality at distances $r \lesssim 1/m_2$. Crucially, the violation is confined to the Compton wavelength of the massive mode: for $\Lambda \sim$ meV, $1/m_2 \sim 100\,\mu m$, far below

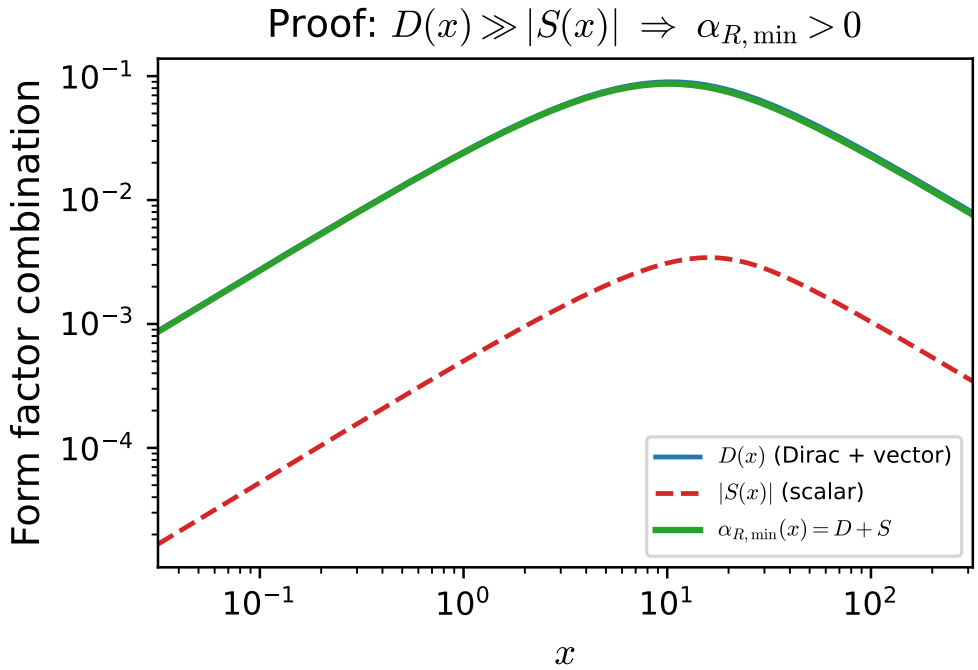

Figure 3: Decomposition of $\alpha_{R,\mathrm{min}}(z) = D(z) + S(z)$ (Eq. 15). The Dirac+vector contribution $D(z)$ (blue) dominates the scalar contribution $|S(z)|$ (red dashed) by a factor > 20 at all z. The resulting $\alpha_{R,\mathrm{min}}$ (green) is strictly positive.

any macroscopic scale. The signal front velocity remains exactly c (the retarded propagator controls the light cone), and the S-matrix remains unitary and Lorentz-covariant order by order in perturbation theory [10]. Macroscopic causality is therefore preserved; only the notion of a sharp microscopic particle trajectory for the spin-2 mode is lost, which is consistent with its absence from the asymptotic spectrum.

This should be contrasted with Stelle gravity [7] (8 helicity states: 2 massless + 5 massive spin-2 + 1 scalar), $f(R)$ gravity (3 DoF, including a scalaron), and Hořava–Lifshitz gravity (3 DoF, including an extra scalar).

In the Anselmi–Piva framework [10], both the scalar and spin-2 modes of generic quadratic gravity can be quantized as fakeons, also yielding 2 physical DoF. The distinction is that in SCT the scalar mode is *absent at the propagator level* (no pole exists), rather than being projected out by prescription. Furthermore, infinite-derivative gravity (IDG) [9] achieves ghost-freedom by postulating an entire-function form factor; in SCT, the form factors are *derived* from the spectral action with SM matter content.

The absence of the scalar mode eliminates the van Dam–Veltman–Zakharov (vDVZ) discontinuity [13, 14]: the SCT propagator smoothly reduces to the GR propagator as $\Lambda \to \infty$, with no extra 1/3 contribution from a scalar exchange.

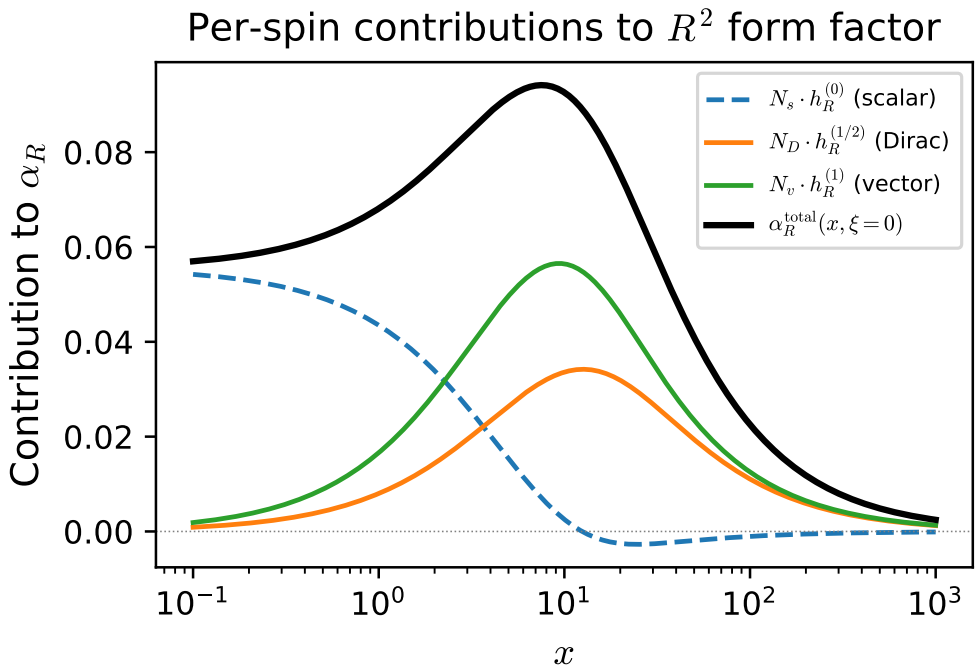

Figure 4: Per-spin contributions to the total R^2 form factor $\alpha_R(z, \xi = 0)$. The scalar contribution (dashed) turns negative for $z \gtrsim 15$, but the Dirac and vector contributions (both always positive) ensure $\alpha_R > 0$.

5 Robustness under matter sector extensions

For a general matter spectrum with N_s real scalars, N_f Weyl fermions, and N_v gauge bosons, the no-scalaron condition $\alpha_{R,\min}(z) > 0$ requires

$$\frac{N_f}{2}\, h_R^{(1/2)}(z) + N_v\, h_R^{(1)}(z) > N_s\, |S_1(z)| \quad \text{for all } z > 0, \tag{20}$$

where $S_1(z) = f_{R,\text{bis}} - f_{RU}^2/(4 f_U)$ is the per-scalar contribution. The critical ratios are:

$$\frac{N_f}{N_s} > 2.37, \quad \frac{N_v}{N_s} > 0.25 \quad \text{(either sufficient)}. \tag{21}$$

The SM values $N_f/N_s = 11.25$ and $N_v/N_s = 3.0$ exceed these thresholds by factors of 4.7 and 12, respectively.

All physically motivated BSM extensions (sterile neutrinos, dark photon, additional scalar singlet) preserve the no-scalaron theorem. Only a *pure scalar* theory ($N_f = N_v = 0$) develops a scalar graviton, a scenario excluded by the observed matter content of the universe. Figure 5 shows the safe region in the $(N_f/N_s,\ N_v/N_s)$ plane.

6 Gravitational wave phenomenology

6.1 Speed and polarization

On FLRW backgrounds, the Weyl tensor vanishes ($C_{\mu\nu\rho\sigma} = 0$), and the tensor perturbation equation gives [12]

$$c_T = c\,\left[1 + \mathcal{O}(H^2/\Lambda^2)\right]. \tag{22}$$

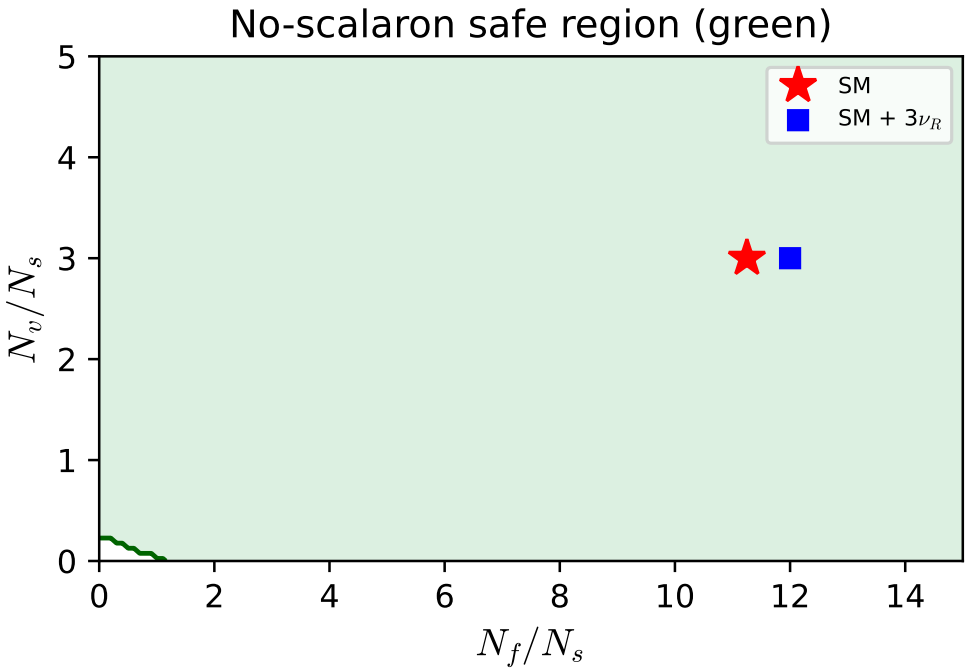

Figure 5: No-scalaron safe region (green) in the matter content plane (N_f/N_s, N_v/N_s). The SM (red star) lies deep inside the safe region; BSM extensions (blue square: SM + 3 sterile neutrinos) remain safe. The boundary is determined by the critical ratios (21).

For $\Lambda \geq 3.53$ meV (torsion-balance bound, Eq. 26), $|c_T/c - 1| \lesssim 10^{-62}$, trivially satisfying the GW170817 constraint $|c_T/c - 1| < 5.6 \times 10^{-16}$ [15].

The two tensor polarizations of GR are the only propagating modes. No scalar breathing mode exists (Theorem 1), providing a falsifiable prediction for scalar polarization searches in LIGO/Virgo/KAGRA and future detectors.

6.2 Dispersion relation

The linearized field equation $k^2\,\Pi_{\rm TT}(k^2/\Lambda^2)\,\tilde{h}_{\mu\nu} = 0$ factorizes into two branches:

$$k^2 = 0 \quad \text{(massless graviton)}, \qquad \Pi_{\rm TT}(k^2/\Lambda^2) = 0 \quad \text{(massive fakeon)}. \tag{23}$$

The physical gravitational wave satisfies $\omega^2 = |\mathbf{k}|^2$ *exactly* on flat backgrounds: the form factor modifies the wave operator but does not shift the massless pole. The massive branch $k^2 = z_0\Lambda^2$ corresponds to the spin-2 fakeon (not an asymptotic state).

On cosmological (FLRW) backgrounds, the Weyl tensor vanishes and the tensor perturbation equation gives curvature-induced corrections $\omega^2 = k^2[1 + \mathcal{O}(H^2/\Lambda^2)]$, as proven by explicit reduction of the full nonlocal field equations to FLRW in Ref. [12] (Theorem 5.1 therein). For $\Lambda \gtrsim 1$ meV, this correction is $\lesssim 10^{-62}$ and completely undetectable.

Remark 5. *The coefficient* 13/60 *in the Taylor expansion of* $\Pi_{\rm TT}$ *appears in the off-shell wave operator at* $\mathcal{O}(k^4/\Lambda^2)$ *but does* not *modify the on-shell dispersion of the physical graviton. A formal mapping to the Mirshekari–Yunes–Will parametrization [16] gives* $A_4^{\rm formal} = 13/(60\Lambda^2)$*, which, using the GWTC-3 bound* $A_4 < 3.0 \times 10^3$ eV^{-2} *[17], yields* $\Lambda > 8.50$ *meV. However, this is an off-shell constraint on the wave operator coefficient, not a bound on physical GW dispersion (which is identically zero in SCT on flat backgrounds).*

6.3 Experimental bounds

The strongest *on-shell* constraint on Λ comes from the dedicated torsion-balance analysis:

Channel	$\Lambda_{\min}$ (meV)	Reference
Torsion balance (Stelle approx.)	2.38	[12]
Torsion balance (SCT dedicated)	**3.53**	this work, Eq. (26)
Off-shell MYW (GWTC-3, formal)	8.50	[17]

The dedicated torsion-balance bound (3.53 meV) is the strongest on-shell constraint. The off-shell MYW bound (8.50 meV) from GWTC-3 constrains the wave operator coefficient but does not correspond to physical GW dispersion (see Remark above).

7 Discussion

7.1 Comparison with other theories

Table 1: Gravitational spectrum of higher-derivative gravity theories.

Theory	Scalar	Spin-2	DoF	UV behavior
GR	—	—	2	non-renorm.
SCT (this work)	**no real pole**	**fakeon**	**2**	**one-loop**
Stelle (1977)	ghost	ghost	8	renorm.
$f(R)$	scalaron	—	3	non-renorm.
IDG	— (postulated)	—	2	finite (postulated)
Hořava–Lifshitz	extra mode	—	3	power-counting

SCT is distinguished from IDG by its *derivation* from first principles: the form factors arise from the spectral action with SM matter content, not from a postulated entire function. The no-scalaron theorem is a *derived consequence*, not an input.

In Brans–Dicke theory [18], the scalar mode decouples only in the strong-coupling limit $\omega \to \infty$; in generic scalar-tensor and Horndeski theories, the scalar is present for all finite parameter values. In $f(R)$ gravity, a propagating scalaron is an unavoidable consequence of the higher-derivative structure. SCT is unique in that the scalar mode is eliminated by the *matter content* (conformal fermions and gauge bosons), not by tuning a gravitational coupling to a special value.

7.2 Perturbative stability

The no-scalaron theorem is derived at one loop. At L loops, the form factors receive corrections proportional to $(G_N\Lambda^2)^{L-1}$. Since $G_N\Lambda^2 = (\Lambda/M_P)^2 \lesssim 10^{-60}$ for Λ at the meV scale, higher-loop corrections to $\alpha_R(z,\xi)$ are suppressed by at least sixty orders of magnitude. Since the one-loop value $\alpha_R \sim 0.05$ is positive by a finite margin (not a near-cancellation), a 10^{-60}-level correction cannot change its sign: no scalar pole can be generated perturbatively. The theorem is therefore perturbatively stable. Non-perturbative effects (gravitational instantons, infrared resummation) are beyond the scope of the present one-loop analysis.

7.3 Gravitational spectroscopy

The results of this paper, combined with the fakeon prescription [5], yield the *real-axis* gravitational particle spectrum of SCT from the SM field content alone (additional complex Lee–Wick poles exist; see the unitarity analysis):

Mode	Mass	Status
Graviton	0	2 polarizations (as GR)
Spin-2 fakeon	$m_2 = 1.554\,\Lambda$	virtual only (0 asymptotic DoF)
Scalar	—	absent (Theorem 1)

The entire spectrum is controlled by a single scale Λ. If Λ is measured from any one channel (GW dispersion, short-range gravity, or astrophysical observations), all other observables are *predicted* with no remaining freedom. This "one-scale spectroscopy" is a direct consequence of the parameter-free nature of the spectral action: the gravitational couplings α_C, α_R, and the spin-2 mass ratio m_2/Λ are all fixed by the SM particle content.

7.4 Modified Newtonian potential

The absence of the scalar mode simplifies the static gravitational potential. In Stelle gravity [7], the Yukawa approximation gives [12]

$$\frac{V_{\text{Stelle}}(r)}{V_N(r)} = 1 - \frac{4}{3}\,\mathrm{e}^{-m_2 r} + \frac{1}{3}\,\mathrm{e}^{-m_0 r}, \tag{24}$$

where $m_0 = \Lambda\sqrt{6}$ is the scalar mass at $\xi = 0$. The spin-2 and scalar Yukawa corrections cancel at $r = 0$, giving $V_{\text{Stelle}}(0) = 0$.

The no-scalaron theorem eliminates the scalar Yukawa (Π_s has no zero), while the spin-2 contribution persists: the static potential is insensitive to the fakeon prescription because the retarded and advanced propagators coincide at zero frequency [10]. Since m_2 depends only on α_C (which is ξ-independent), the SCT potential for all ξ reduces to

$$\boxed{\frac{V_{\text{SCT}}(r)}{V_N(r)} = 1 - \frac{4}{3}\,\mathrm{e}^{-m_2 r},} \tag{25}$$

a single-Yukawa correction with $m_2 = 1.554\,\Lambda$. This is simpler than the Stelle potential (24) and qualitatively distinct:

- $V_{\text{SCT}}(0)/V_N = -1/3$ in the Yukawa approximation, vs. $V_{\text{Stelle}}(0)/V_N = 0$ (exact cancellation);
- no attractive scalar tail at intermediate r;
- monotonic approach to V_N from above for $r \gtrsim 1/m_2$ (Fig. 6).

We stress that both (24) and (25) are *local* (Yukawa) approximations, valid for $r \gg 1/\Lambda$. At $r \lesssim 1/\Lambda$, the full nonlocal form factors dominate and the pointwise values $V(0)$ are not physical. In particular, the $V_{\text{Stelle}}(0) = 0$ cancellation is also an artifact of the local approximation, not a genuine regularization of the Newtonian singularity [7].

The absence of the scalar Yukawa has a clear physical consequence: the partial cancellation between spin-2 and scalar corrections that softens the potential at short distances in

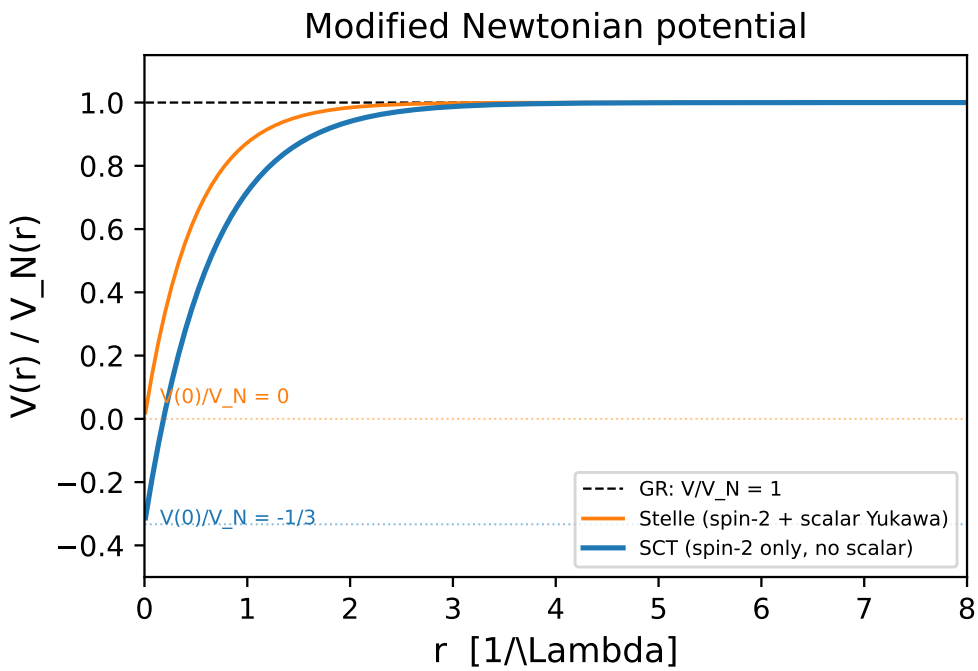

Figure 6: Modified Newtonian potential $V(r)/V_N(r)$. Stelle gravity (orange) has both spin-2 and scalar Yukawa corrections that cancel at $r = 0$. SCT (blue) has only the spin-2 Yukawa (Theorem 1 eliminates the scalar), producing a repulsive core $V(0)/V_N = -1/3$.

Stelle gravity does not occur in SCT. Within the local approximation, the SCT potential retains a $1/r$ singularity (with a repulsive sign). This means that the no-scalaron theorem, while eliminating the scalar degree of freedom, does *not* by itself resolve the point-source singularity. Short-distance regularization in SCT must come from a different mechanism: the full nonlocal structure of the form factors at $r \sim 1/\Lambda$, or from a nonperturbative UV completion (which is beyond the scope of the present one-loop analysis). This situation is analogous to QED, where the one-loop vacuum polarization modifies but does not remove the Coulomb singularity; the resolution requires the full nonperturbative theory.

7.5 Falsifiability and experimental signatures

The no-scalaron theorem leads to three distinct experimental signatures:

Scalar gravitational wave polarization. SCT predicts zero scalar breathing-mode amplitude for any ξ. Detection of a scalar GW polarization in LIGO/Virgo/KAGRA or future detectors (ET, CE) would rule out SCT. Current O3 bounds constrain the scalar-to-tensor ratio to $\mathcal{O}(1)$; third-generation detectors will reach $\mathcal{O}(10^{-2})$.

Dedicated torsion-balance bound. The SCT potential (25) is a single-Yukawa correction with amplitude $|\alpha| = 4/3$ and range $\lambda = 1/m_2 = 1/(1.554\,\Lambda)$. Using the Lee et al. (2020) [19] exclusion curve at $|\alpha| = 4/3$, the 95% CL bound is $\lambda < 36.0\,\mu$m, giving

$$\Lambda > \frac{\hbar c}{1.554 \times 36.0\,\mu\text{m}} = 3.53 \text{ meV} \quad \text{(torsion balance, dedicated).} \tag{26}$$

This is 1.5× stronger than the bound derived from the double-Yukawa (Stelle) parametrization ($\Lambda > 2.38$ meV) in Ref. [12], owing to the exact spin-2 mass ($m_2 = 1.554\,\Lambda$ vs. $2.148\,\Lambda$ locally) and the correct signal amplitude ($|\alpha| = 4/3$, not rescaled from the $|\alpha| = 1$ benchmark).

Caveat: the Yukawa range $1/m_2 = 0.64/\Lambda$ is comparable to $1/\Lambda$, the scale at which the full nonlocal propagator deviates from the local approximation. The bound (26) should therefore be regarded as an order-of-magnitude estimate; a precise constraint requires the full nonlocal potential, which we defer to future work.

Spectral action scale Λ from multiple channels. If Λ is measured from GW dispersion (GWTC-3 or future runs) and independently from short-range gravity, the two values must agree. The single parameter Λ controls both channels with no freedom to adjust: inconsistency would falsify SCT.

8 Conclusion

We have proven that spectral causal theory propagates exactly two gravitational degrees of freedom, the same as general relativity. The scalar gravitational mode, which is generically present in higher-derivative gravity, is universally absent due to the positive contributions of conformal matter (fermions and gauge bosons) to the R^2 form factor at nonzero momentum.

The key results are:

1. **No-scalaron theorem** (Theorem 1): $\alpha_R(z,\xi) > 0$ and $\Pi_s(z,\xi) > 1$ for all $z > 0$ and all $\xi \in \mathbb{R}$, including conformal coupling. No real scalar mass pole exists. This is a parameter-free prediction from the SM spectrum.

2. **Exact spin-2 mass:** $m_2 = 1.554\,\Lambda$, a 38% reduction from the local approximation ($m_2^{\rm loc} = 2.148\,\Lambda$).

3. **Torsion-balance bound:** $\Lambda > 3.53$ meV (dedicated single-Yukawa analysis, 1.5× stronger than the Stelle parametrization).

4. **Robustness:** the SM lies 4.7× (fermion) and 12× (vector) above the critical thresholds; all BSM extensions preserve the theorem.

5. **Falsifiability:** detection of scalar GW polarization at any frequency would rule out SCT.

Use of AI tools. Large language models (Claude, Anthropic) were used for code generation and numerical verification scripting. All mathematical derivations, physical arguments, and scientific conclusions were formulated and verified by the authors. The AI-generated code was independently validated against analytical results at 100-digit precision.

Data availability. All numerical computations are reproducible from the publicly available code repository [20].

A Explicit form factor expressions

All form factors are expressed in terms of the master function $\varphi(z) = \int_0^1 \mathrm{d}\alpha\, \mathrm{e}^{-z\alpha(1-\alpha)} = \mathrm{e}^{-z/4}\sqrt{\pi/z}\,\mathrm{erfi}(\sqrt{z}/2)$, with $\varphi(0) = 1$, $\varphi'(0) = -1/6$.

Dirac (spin-$\frac{1}{2}$).

$$h_R^{(1/2)}(z) = \frac{3\varphi + 2}{36\,z} + \frac{5(\varphi - 1)}{6\,z^2}, \quad h_R^{(1/2)}(0) = 0. \tag{27}$$

Vector (spin-1).

$$h_R^{(1)}(z) = -\frac{\varphi}{48} + \frac{11 - 6\varphi}{72\,z} + \frac{5(\varphi - 1)}{12\,z^2}, \quad h_R^{(1)}(0) = 0. \tag{28}$$

Scalar (spin-0). The scalar form factor decomposes as $h_R^{(0)}(z,\xi) = f_{R,\mathrm{bis}}(z) + \xi\, f_{RU}(z) + \xi^2 f_U(z)$, where

$$f_U(z) = \varphi/2, \tag{29}$$

$$f_{RU}(z) = -\varphi/4 - (\varphi - 1)/(2z), \tag{30}$$

$$f_{R,\mathrm{bis}}(z) = \frac{1}{18\,z} + \frac{\varphi}{32} + \frac{\varphi}{8\,z} - \frac{7}{48\,z} - \frac{\varphi - 1}{8\,z^2} + \frac{\varphi - 1}{3\,z^2}. \tag{31}$$

The per-scalar contribution to the minimum is $S_1(z) = f_{R,\mathrm{bis}} - f_{RU}^2/(4\,f_U)$, with UV asymptotics $z\,S_1 \to -1/36$ and $z\,D \to 31/12$, giving $|S/D|_{z\to\infty} = 4/93 \approx 0.043$.

B Derivation of the integral representations

Define moments $I_n(z) = \int_0^1 \tau^n \mathrm{e}^{-z\tau}\,\mathrm{d}\alpha$ with $\tau = \alpha(1-\alpha)$, so $I_0 = \varphi$.

Integration by parts using $\tau'(\alpha) = 1 - 2\alpha$ and $(1-2\alpha)^2 = 1 - 4\tau$ yields

$$4z\,I_1 = (z+2)\,\varphi - 2, \tag{32}$$

and for $n \geq 1$ the recurrence

$$4z\,I_{n+1} = (z + 4n + 2)\,I_n - n\,I_{n-1}. \tag{33}$$

Proof of (17). From (32) and (33), compute I_2 and I_3 explicitly. Direct algebra then gives $36z^2\,h_R^{(1/2)} = z^3(I_1 - 8I_2 + 16I_3)$. Since $\tau - 8\tau^2 + 16\tau^3 = \tau(1-4\tau)^2 \geq 0$ (with $(1-4\tau) = (1-2\alpha)^2 \geq 0$), the integrand is non-negative, establishing (17).

Proof of (18). Similarly, $144z^2\,h_R^{(1)} = 8z^3(I_1 - 5I_2 + 4I_3)$. The kernel $\tau - 5\tau^2 + 4\tau^3 = \tau(1-\tau)(1-4\tau)$ is non-negative on $\tau \in [0, 1/4]$, establishing (18).

C Uniqueness of the spin-2 mass

Proposition 6. $\Pi_{\rm TT}(z)$ *has exactly one positive real zero* $z_0 \in (2, 5/2)$.

Proof sketch. Step 1: $\Pi_{\rm TT}(z) > 0$ on $(0, 2]$. Expand φ to fourth order and bound $\Pi_{\rm TT}$ below by a polynomial $P_-(z) = 1 + \frac{13}{60}z - \frac{307}{840}z^2 + \frac{5}{112}z^3 - \frac{1}{140}z^4$, which is concave ($P_-'' < 0$ everywhere, since discriminant < 0) and satisfies $P_-(0) = 1$, $P_-(2) = 3/14 > 0$.

Step 2: $\Pi_{\rm TT}(5/2) < 0$ from the upper Taylor bound.

Step 3: At any positive zero z_*, expressing $\varphi(z_*)$ from $\Pi_{\rm TT}(z_*) = 0$ and using $4z\varphi' + (z+2)\varphi = 2$ (from (32)) yields

$$\Pi'_{\rm TT}(z_*) = -\frac{996z_*^3 + 5799z_*^2 - 5600z_* - 8496}{24z_*(12z_*^2 + 93z_* + 236)}. \tag{34}$$

For $z_* \geq 2$, the numerator factors as $(z_* - 2)(996z_*^2 + 7791z_* + 9982) + 11468 > 0$, so $\Pi'_{\rm TT}(z_*) < 0$. A second zero would require $\Pi_{\rm TT}$ to return to zero from below, forcing $\Pi'_{\rm TT}(z_2) \geq 0$, contradicting $\Pi'_{\rm TT}(z_2) < 0$. □

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